

d) Aim: Develop an application that demonstrates screen navigation using named routes and `Navigator.pushNamed()`.

Description:

- **MyApp:** The main widget of the application. It uses `MaterialApp` with named routes.
 - The `initialRoute` is set to `'/'`, which opens the **Home Screen**.
 - Three routes are defined:
 - `'/'` → **Home Screen**
 - `'/second'` → **Second Screen**
 - `'/third'` → **Third Screen**
- **Home Screen:** Contains:
 - An `AppBar` titled **“Home Screen”**.
 - A button labeled **“Go to Second Screen”**.
 - The button uses `Navigator.pushNamed()` to navigate to `'/second'`.
- **Second Screen:** Contains:
 - An `AppBar` titled **“Second Screen”**.
 - A button labeled **“Go to Third Screen”**, which uses `Navigator.pushNamed()` to navigate to `'/third'`.
 - A button labeled **“Go Back to Home”**, which uses `Navigator.pop()` to return to the previous screen.
- **Third Screen:** Contains:
 - An `AppBar` titled **“Third Screen”**.
 - A button labeled **“Go Back to Second Screen”**, which uses `Navigator.pop()` to return to the Second Screen.
 - A button labeled **“Go Back to Home”**, which uses `Navigator.pushNamed()` to navigate to the Home Screen using the `'/'` route.

Source Code:

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
// Main App
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Data Passing Demo',
```

```
        home: const HomeScreen(),
    );
}
}
// Home Screen
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Home Screen'),
      ),
      body: Center(
        child: ElevatedButton(
          child: const Text('Go to Detail Screen'),
          onPressed: () {
            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) {
                  return const DetailScreen(
                    data: 'Hello from Home!',
                  );
                },
              ),
            );
          },
        ),
      ),
    );
  }
}
// Detail Screen
class DetailScreen extends StatelessWidget {
  final String data;
  const DetailScreen({
    super.key,
    required this.data,
  });
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Detail Screen'),
      ),
      body: Center(
        child: Text(
          data,
          style: const TextStyle(
            fontSize: 24,
```

Date:

Week-04

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```
fontWeight: FontWeight.bold,  
),  
),  
),  
);  
}  
}
```

Output:

